DO NOW — RECALL

Scope, modularity and parameters

Name: _____ Class: _____ Date: _____

1. If procedural Deposit(balance,5) returns a value that the caller ignores, does balance change?

2. What is a local variable?

3. How does a value-returning function pass a result back?

4. Why might using ref unexpectedly make a method harder to debug?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. The returned value is discarded.

2. Model answer

A variable declared inside a method or block, with restricted scope.

3. Model answer

Using return.

4. Model answer

The method can change a caller's variable as a side effect, even if there is no returned result.

Vocabulary — know these terms

scope — Where a name can be accessed.

local — Variable with block or method-level scope.

global/class-level — Variable accessible in a wider scope, subject to access rules.

by value — Pass a copy of the value.

by reference — Pass an alias of the caller's variable.

NEW LEARNING

Read, annotate and explain

Copying an integer

Assignment b=a copies the integer 5, not a reference to a storage location.

Passing by reference

Increase(ref a) gives parameter value an alias of a, so the caller a changes to 8.

Use modular methods

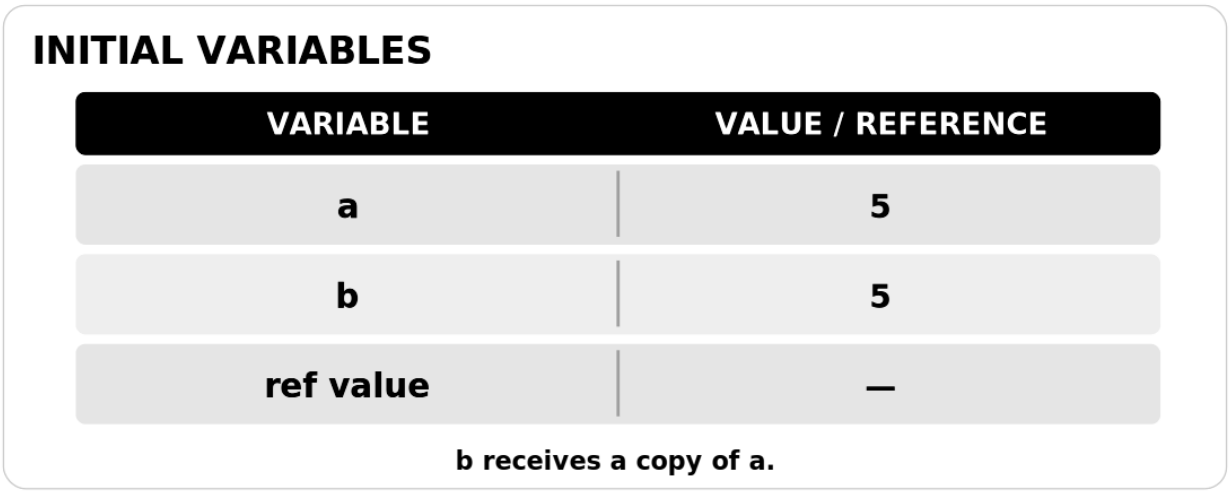
A short reusable method has a clear input effect, can be tested, and reduces repeated code.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

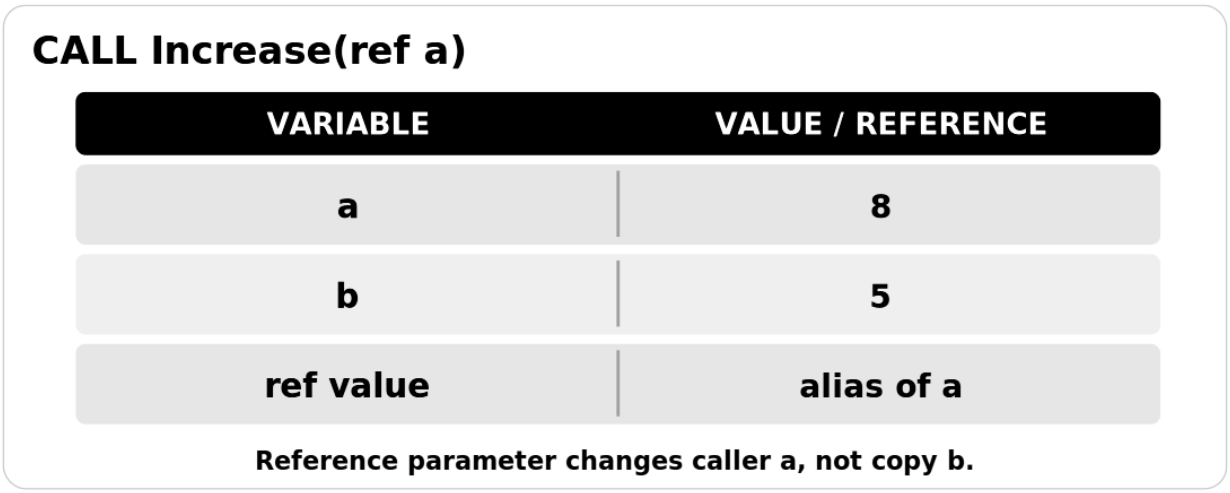
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INITIAL VARIABLES: a=5, b=5, ref value=—. b receives a copy of a.

New Learning



CALL Increase(ref a): a=8, b=5, ref value=alias of a. Reference parameter changes caller a, not copy b.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

AFTER RETURN

VARIABLE	VALUE / REFERENCE
a	8
b	5
ref value	out of scope

a and b now differ.

AFTER RETURN: a=8, b=5, ref value=out of scope. a and b now differ.

I DO — WORKED EXAMPLE

Follow the worked method

Trace AddFour(x) by value when caller x starts at 6.

Predict first

The function parameter number receives a copy of 6.

Model and check

Inside the function number becomes 10; caller x remains 6.

Model and check

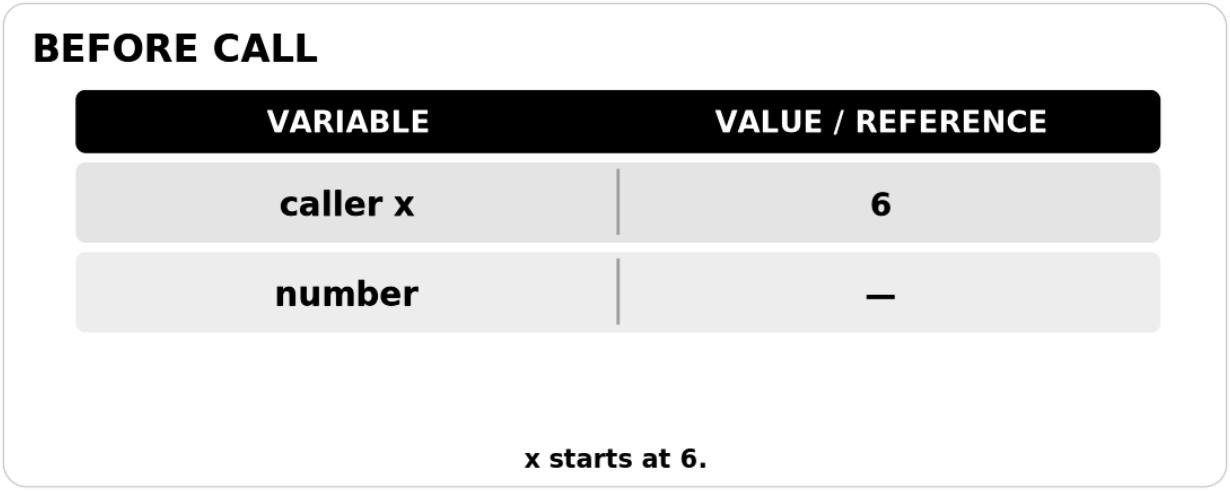
The parameter leaves scope at the end of the call.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace AddFour(x) by value when caller x starts at 6.

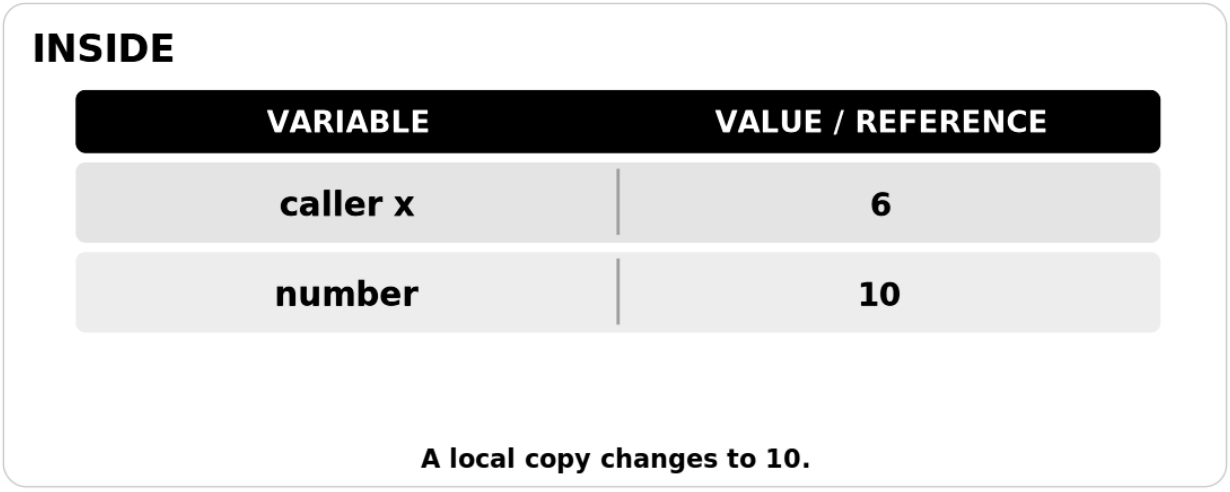
The function parameter number receives a copy of 6.



BEFORE CALL: caller x=6, number=—. x starts at 6.

Trace AddFour(x) by value when caller x starts at 6.

The function parameter number receives a copy of 6.



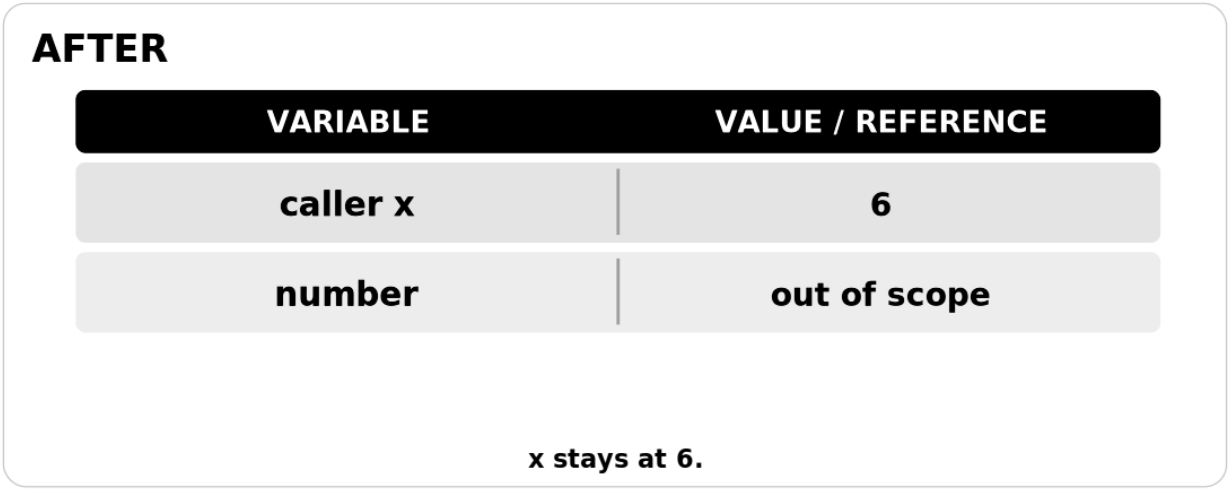
INSIDE: caller x=6, number=10. A local copy changes to 10.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace AddFour(x) by value when caller x starts at 6.

The function parameter number receives a copy of 6.



AFTER: caller x=6, number=out of scope. x stays at 6.

WE DO — GUIDED PRACTICE

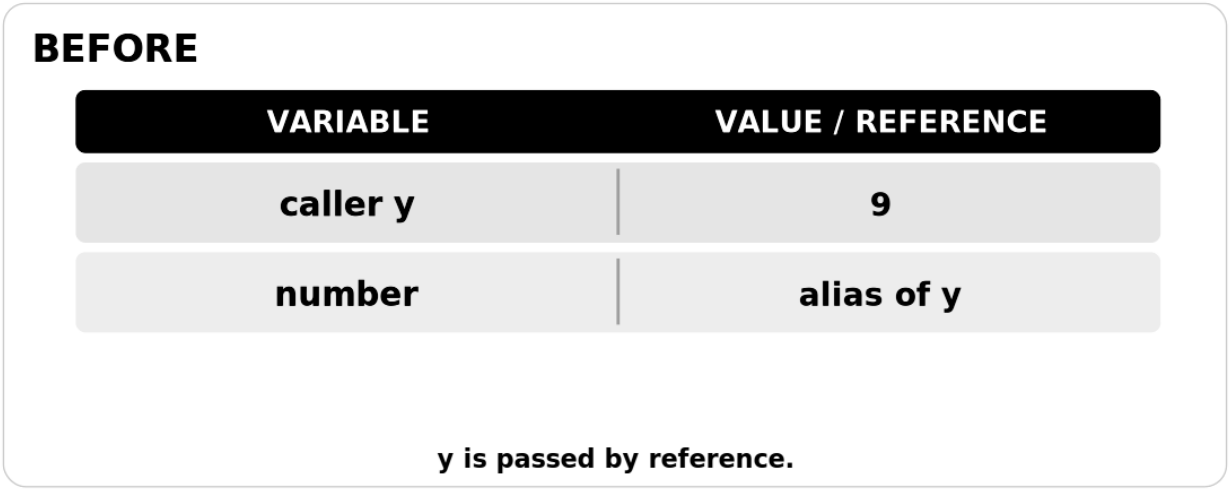
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace AddFour(ref y) when caller y starts at 9.

The ref parameter number refers to y, not a fresh copy.



BEFORE: caller y=9, number=alias of y. y is passed by reference.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The ref parameter number refers to y, not a fresh copy.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Method AddTwo(int n) adds 2 to n. Caller x=7 is passed by value. What is x after the call?

Question 2

Write an OCR-style subroutine Swap with two by-reference integer parameters and explain why by-reference is required.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare `int result=Double(x);` with `Double(ref x);` as ways of updating a number. Give advantages of one method for a pure calculation.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Passing an integer by value in C# changes the caller's integer when the parameter changes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

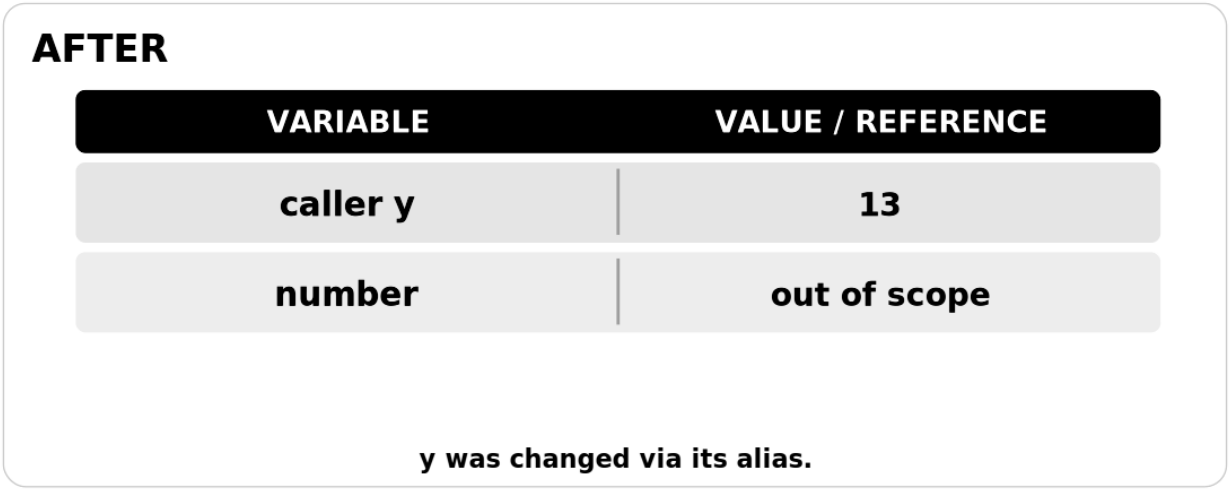
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace AddFour(ref y) when caller y starts at 9.

The ref parameter number refers to y, not a fresh copy.



AFTER: caller y=13, number=out of scope. y was changed via its alias.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Adding four changes caller y from 9 to 13.

IMPROVE: Explain one correction to your own reasoning.